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Research Article

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Posted Date: July 2nd, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-9903352/v1>

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Additional Declarations: No competing interests reported.

Optimization Algorithms of the Proposed FOC of Permanent Magnet Synchronous Motor for Electric Vehicles

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Abstract: The optimization algorithms used for the suggested Field-Oriented Control FOC strategy of a Permanent Magnet Synchronous Motor PMSM for electric vehicle applications. The goal of the study is to improve the PMSM drive system's stability, efficiency, and dynamic performance across a range of operating scenarios. The creation of a comprehensive electric vehicle model took into account the electric motor, powertrain, and vehicle dynamics. To determine how vehicle mass affected performance and energy consumption, three different vehicle weights, 600 kg, 900 kg, and 1200 kg, were simulated. The FOC system incorporates advanced optimization approaches to minimize current ripple and electromagnetic torque variations while improving speed tracking, torque response, and energy utilization. MATLAB Simulink is used to design and simulate the PMSM model and control system in order to assess the efficacy of the suggested methods. The refined FOC technique achieves greater performance in terms of quick transient response, decreased steady-state error, and enhanced overall drive efficiency, according to comparative simulation results, making it ideal for contemporary electric vehicle propulsion systems.

Key words: Battery electric vehicle (BEV), Fuzzy Proportional-Integral-Derivative controller, Brushless DC, switched reluctance motors, Field-oriented control (FOC).

1. Introduction:

The demand for high-performance motor drive systems that can offer excellent efficiency, dependability, and precise speed control has surged due to the quick expansion of electric vehicles EVs. Because of its high power density, exceptional efficiency, small size, and outstanding dynamic response, the Permanent Magnet Synchronous Motor PMSM has emerged as one of the most popular options among the different electric motor technologies for EV propulsion systems. In order to effectively benefit from PMSM, sophisticated control systems are needed to provide precise torque and speed management under various operating situations. Because it allows for independent control of torque and flux components, much like individually excited DC motors, Field-Oriented Control FOC is commonly utilized in PMSM drive systems. This management strategy offers smooth motor running with less torque ripple and greatly enhances dynamic performance. Conventional FOC methods, however, may have drawbacks such lower performance under abrupt load fluctuations and nonlinear operating circumstances, steady-state errors, and parameter sensitivity. In order to improve controller accuracy, robustness, and overall efficiency, optimization methods are being used with FOC systems. [1] was to construct and evaluate an in-wheel outer-rotor permanent magnet synchronous motor (PMSM) for electric vehicles, using a previously developed model. While maintaining the same dimensions and volume of the chosen machine, an optimization process is used to increase the torque produced. The finite element

software package "MagNet 7.4.1," the Visual Basic 16.0 programming language, and MATLAB 9.5 Simulink for post processing are used to build and test the proposed machine.

Used metamodels to design the geometry of a PMSM to reduce torque ripple, thereby improving the motor's vibration and noise performance. Additionally, a sequential two-point diagonal quadratic approximation optimization procedure was used to combine the optimal metamodels for every variable. While the constraint settings remained the same, the torque ripple improved by about 1.63% compared with the original model. Torque control methods for interior permanent magnet synchronous motors PMSM in electric vehicles were thoroughly examined and compared [3]. Maximum torque per ampere, minimum torque ripple, maximum efficiency, quick dynamics, and a broad speed range are among the vehicle requirements that the control seeks to satisfy. Iron losses are accounted for by the PMSM model, ensuring accurate results. To precisely evaluate the overall system efficiency, the IGBT and diode losses (conduction and switching) in the power inverter are also modeled and computed.

A comprehensive method for limited control of a permanent magnet synchronous machine was provided [4]. It makes use of proportional-integral current controllers adjusted to achieve zero steady-state error and a quick transient response through field-oriented control. The suggested approach is tested on a surface-mounted permanent magnet synchronous machine and implemented on a Texas Instruments F28343 200 MHz microcontroller. [5] Through analytical evaluations and practical investigations of the motor utilized in a commercial two-wheeled LEV, the specifications and dimensions of the PMSM have been established in this work. Four magnet configurations V-shape, U-shape, double V-shape, and double I-shape were assessed using electromagnetic studies; the double I-shape arrangement was chosen due to its higher performance. ANSYS OptiSlang was then used to conduct optimization studies to improve the motor's overall efficiency and design, guided by specified cost functions. Furthermore, the Double-I structure improves manufacturability while retaining similar thermal performance as compared to the delta-type reference PM rotor geometry.

An efficiency optimization controller for an electric vehicle's permanent magnet synchronous motor (PMSM) was introduced [6]. Based on the new loss model, a novel loss minimization algorithm (LMA) is created by modifying the excitation current ratio on the d-q axis. Additionally, the suggested approach makes use of the Hamiltonian system's energy shape method. The primary findings of these simulations show that the drive's dynamic performance is preserved and that efficiency can be increased by up to 7% when compared to the $i_d = 0$ control scheme. Important processes in the machine's design optimization, analysis, construction, and testing were detailed [7]. Design optimization achieved a torque and efficiency of 20 Nm and 98.6%, respectively, by using a hybrid analytical/finite element model in conjunction with the non-dominated sorting genetic algorithm II (NSGA-II) to minimize computing time. A prototype has been built. Reliability and motor temperatures with and without water refrigeration at nominal current were assessed by the test results. The results show that there was little effect on the vehicle's lap time, which went from 77.3 seconds for the ideal optimized motor to 78.9 seconds for the prototyped motor.

Using a fuzzy logic controller for electric vehicles, [8] described the control design of an in-wheel axial-flux permanent magnet synchronous motor with one stator and one rotor. A 5x5 matrix was used to create the fuzzy logic controller to satisfy the controller's necessary output stator voltage. The PI controller and the fuzzy logic torque controller were contrasted in terms of speed, torque, and stator current responsiveness. MATLAB/SIMULINK simulation results were used to assess the suggested controller.

An approach to creating a cascade control scheme for a permanent magnet synchronous motor (PMSM) used in electric vehicles (EVs) was given [9]. A Proportional Integral (PI) controller for the inner loop current control is based on the constructed dynamic model of the PMSM drive system by taking input-output limitations into account, while Model Predictive Control (MPC) is devised for the outer loop speed control. The proportional integral gain's parameters and the cost function's weighting matrix are adjusted using the genetic algorithm technique. According to the simulation results, the suggested MPC outperformed LQR and GA-based PI in terms of performance and effectiveness. To optimize the control of permanent magnet synchronous motors PMSM [10] introduced a new model predictive current control (MPCC) method that was combined with the adaptive neuro-fuzzy inference system (ANFIS) and called ANFIS-MPCC. The suggested ANFIS-MPCC approach is contrasted with conventional control techniques like proportional-integral MPCC (PI-MPCC) and artificial neural network MPCC (ANN-MPCC) to assess how well it addresses problems like overshoot reduction, settling time minimization, and harmonic distortion (THD) suppression. With overshoot decreased to 0.015%, a settling time of 0.00147 seconds, and THD minimized to 2.0% at rated speed and 2.02% at low speed, the results demonstrate that ANFIS-MPCC works noticeably better than the conventional approaches.

The field-oriented control technique has been applied to a permanent magnet synchronous motor speed controller [11]. The results demonstrate that the Tree-Seed Algorithm is a useful tool for the controller tuning process since it is utilized for the tuning of the controller parameters. To compare, the particle swarm algorithm also adjusts the controller. The findings demonstrate that the tree-seed algorithm-optimized system performs well for applications requiring varying load torque and speed. The optimization of high speed PMSM is essential for its use in FEVs, according to Mahmoud Ibrahim et al. [12]. A simulation model of the motor drive system was created using MATLAB/Simulink. In order to push the limits of motor performance while managing the intrinsic constraints of high-speed operation, the suggested approach combines the concepts of maximum torque per ampere (MTPA) with the flux weakening control technique rather than the traditional zero direct axis current (ZDAC). The results of the suggested method show that the customized control approach greatly improves motor torque performance by over 22%, which is a significant development in the field of high-speed PMSM.

To solve this problem, [13] suggested an intelligent online estimating technique that makes use of the Bacterial Foraging Optimization (BFO) algorithm and Popov hyper stability theory. The suggested method simultaneously assesses the real motor speed and three important PMSM parameters: stator resistance, inductance, and permanent magnet flux. The outcomes demonstrate how well it improves PMSM control performance, which makes it a viable option for high-precision motor applications. When compared to conventional manual tuning, experimental results

show a 12% increase in estimating precision. The effectiveness of an EV-PMSM control technique based on the DT idea was assessed by Mahmoud Ibrahim et al. [14]. The physical, virtual, and service unit (data exchange) components of the PMSM DT architecture were suggested. The main purpose of EV-PMSM DT was to assess the motor control algorithm. The motor's estimated speed based on its DT was compared to its actual speed. To create a speed estimator that would maximize the motor control approach, the acquired data demonstrated the efficacy of the suggested DT in estimating the motor speed.

To address common issues, including decreased torque ripples, efficiency optimization, and simpler control algorithms, [15] examined PMSM control systems that have been created and are being developed in recent years. As a result, the study examines the most advanced drives, paying special emphasis to reduced model dependency, increased drive range, simpler control methods, and smoother output torque. Improving the dynamic response in the speed profile for various driving conditions is crucial for EVs, according to research [16]. This was accomplished by utilizing the finite control set model predictive control (FCS-MPC) algorithm for PMSM and by using two case studies to compare and assess the control techniques of a PMSM utilized in an EV. OPAL-RT is used to validate the outcomes in the hardware-in-the-loop approach. To increase the efficiency of EVs, [17] concentrated on using Maximum Torque Per Ampere (MTPA) technology in Field-Oriented Control (FOC) of PMSM drives. MATLAB/Simulink is used to evaluate performance, evaluating characteristics such as torque vs. time, KWH vs. time, speed vs. time, and current vs. time. By offering a thorough approach for boosting efficiency, it seeks to improve EV technology and ease the shift to sustainable transportation options.

A novel optimal design method for a permanent magnet synchronous motor (PMSM) in hybrid electric vehicles (HEVs) was presented by Dong Wei et al. [18]. To return and characterize the optimal PMSM design in the case under study, a combinatorial optimization technique is developed to reduce energy loss. An HEV dynamics model has been used to demonstrate and validate the approach's performance. The findings demonstrate that, compared with the original design, the best design method can reduce motor energy losses by 18.35% and increase HEV fuel economy by 3.2% over the driving cycle. [19] proposed an interior permanent magnet (IPM) synchronous machine with multi flux barriers to meet the wide-speed-regulation application requirements of electric vehicles. Based on the flux-barrier characteristics, an electromagnetic-mechanical coupling optimization strategy is employed in the machine design. After machine optimization, comparative investigations are conducted using finite-element analysis, considering both electromagnetic and mechanical performance. [20] reported experimental testing of a spoke-type PMSM developed to meet the requirements of the L6e European light vehicle class. A test bench was developed for this purpose, using a DC machine fed by a bidirectional DC power supply, enabling testing of the PMSM prototype in both motor and generator/brake modes. The test results validated the FEM-simulated results and provided insight into the efficiency of the entire drive system (battery-inverter-PMSM) and the autonomy of the L6e light EV. [21] proposed a multi objective optimization method for an interior permanent magnet synchronous motor for a traction drive PMSM. Based on the flux at wire-winding technology, the multi-objective optimization design of the PMSM is carried out to improve motor power density and high-efficiency range, reduce torque ripple, and suppress electromagnetic vibration and noise. A

mathematical model that accounts for iron losses is established, and the optimization objectives are defined. [22] determined the requirements and dimensions of the PMSM through analytical analyses and experimental studies on a motor used in a commercial two-wheeled LEV. Optimization studies, guided by defined cost functions, were conducted in ANSYS OptiSlang to enhance the motor's design and overall efficiency. In addition, compared to the delta-type reference PM rotor geometry, the Double-I structure enhances manufacturability while maintaining comparable thermal performance.

Proposed a modular vector control scheme for multi-three-phase permanent magnet synchronous motors. The proposed solution uses a modular modeling approach to independently and decoupled control the torque of each three-phase unit [23], enabling torque-sharing strategies among the machine's three-phase sets. The developed modular control has been validated on a nine-phase permanent magnet machine. [24] introduced a novel interior permanent magnet synchronous machine PMSM with a parallel configuration of hybrid rare-earth and ferrite permanent magnets (HPMs) and a V-shaped spoke arrangement. In addition, the electromagnetic performance, mechanical strength at high speeds, and demagnetization resistance of ferrite magnets are analyzed using finite element analysis (FEA). A scaled prototype is built and tested to validate the FEA results. Based on EV operating conditions, Zhao et al. [29] created a reverse-salient PMSM with a high power factor at high speeds and a broad constant-power speed range.

The multipolar PMSM with an uneven tooth arrangement with the same number of poles and slots was created in [30]. The intrinsic cogging torque has been decreased by using this structure of poles and slots of the same number. In order to maximize the flux linkage of the stator teeth and winding synthesis coefficient and, consequently, the electromagnetic torque, the uneven teeth structure was examined. The shape optimization of a PMSM for e-mobility was carried out in this study. The average torque and efficiency were set as restrictions, and the optimal design's objective function sought to reduce the torque ripple. To meet the limitations and the goal function, three design variables were chosen. For the sampling strategy for DoE, there are several approaches, including Plackett-Burman, central composite design, and complete factorial [31].

A LEV's electric drive system is made up of crucial parts that influence how the vehicle moves and functions. Among these, the electric motor is one of the most important parts since it gives the vehicle the torque and propulsion power it needs. These vehicles often use an internal permanent magnet synchronous motor (PMSM) as its electric motor architecture. This is mostly due to the topology's great power density and efficiency as well as its ability to deliver steady power over a broad speed range. Additionally, the IPMSM's construction, which offers a reluctance torque, is a major benefit [32]. A fully functional high-performance drive system with an internal permanent magnet synchronous machine was initially described. Nevertheless, this approach heavily depends on motor characteristics and necessitates precise flux linkage estimate. A formula calculation approach was provided by S. Morimoto et al. to maximize PMSM efficiency [33]. Cooper loss, mechanical loss, and iron loss are some of the motor losses associated with PMSM. For high efficiency, these losses must be reduced, and numerous research concentrate on optimal motor design [34], [35]. The performance of the suggested field-oriented control system for permanent magnet synchronous motors used in electric vehicles is greatly enhanced by optimization methods.

Under various operating situations, these algorithms are used to optimize control parameters like torque response, speed regulation, flux control, and energy economy. The PMSM drive system can deliver faster dynamic response, less torque ripple, better stability, and improved overall vehicle performance by combining sophisticated optimization techniques with the FOC strategy. Additionally, the optimization method helps to increase battery utilization and reduce power losses, both of which are crucial for increasing the driving range of electric vehicles. Optimized The proposed PSO- IFOC based PMSM drives offer better control precision and energy efficiency than traditional control methods, according to simulation and experimental evaluations, making them ideal for contemporary electric car applications.

2. Intelligent Optimization Algorithms (IOA): Particle Swarm

Inspired by the collective social behavior of animals, including fish populations or flocks of birds, the Particle Swarm Optimization Algorithm (PSO) is a population-based, random optimization algorithm. This method, which was created in 1995 by James Kennedy and Russell Eberhart, is essential to swarm intelligence. It is intended to tackle challenging optimization issues by locating the ideal answer (global optimal solution) inside a sizable, nonlinear research space. Fundamental idea and modes of action: Particles: A collection of potential solutions known as "particles" are used by the algorithm. Behavior: Each particle "flies" around a multidimensional search space, modifying its location according to the swarm's overall experience (global best solution, gbest) and its own experience (personal best solution, pbest). Velocity and Location Updates: Particles modify their velocity (V) and position (X) in each iteration using straightforward mathematical formulas that strike a balance between exploring new regions and taking advantage of good known solutions. Iterative optimization: The process is repeated until a satisfactory, optimal, or near-optimal solution is found. Features: Function Independence: The Particle Swarm Optimization (PSO) algorithm does not require the target function to be continuous or differentiable, making it very flexible for solving complex problems. Few Transactions: It is a popular choice in machine learning for feature selection, data collection, and hyper parameter optimization. Iterative optimization: Until a good, ideal, or nearly ideal solution is discovered, the procedure is repeated. Features: Function Independence: The Particle Swarm Optimization (PSO) algorithm is highly adaptable for handling complicated problems since it does not require the target function to be continuous or differentiable as shown in figure 1. Few Transactions: It just takes a few transactions to modify, and it is well known for being straightforward and simple to use. Efficiency: It is a popular option in machine learning for feature selection, data collecting, and hyper parameter optimization since it can successfully navigate across vast and complicated search spaces. Applications: The particle swarm optimization approach has been effectively used in several domains, such as Designing intricate structures and systems is known as engineering optimization. Machine learning: enhancing neural network weights and data classification. Power systems: enhancing operations, scheduling, and power distribution. Feature extraction and picture recording are examples of image processing. Variations and Expansions: Since the particle swarm optimization algorithm's inception, numerous modifications have been made to enhance its performance. These include the addition of inertial weights or the use of various communication topics, like ring or star networks, to enhance mobility and prevent premature convergence.

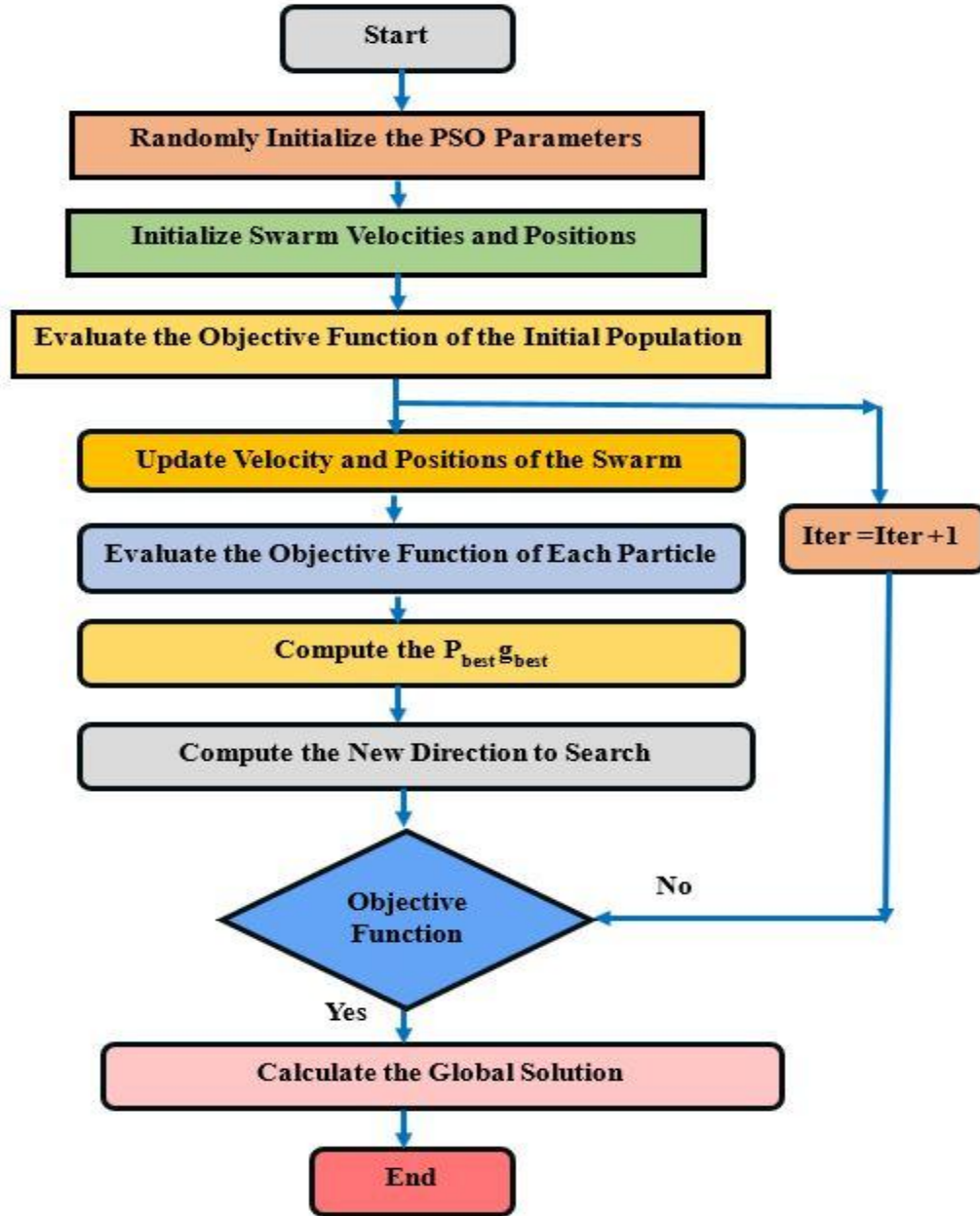


Figure 1: Intelligent Optimization Algorithms PSO- FOC PMSM flowchart

3. Mathematical Vehicle Model

A force balance along the vehicle longitudinal axis yields as shown in figure 2, and electric vehicle dynamics variables as shown in table 1. A mathematical model for the vehicle was constructed in three steps using MATLAB Simulink and mathematical equations.

3.1 Vehicle Model

3.1.1 Resistive Forces to Motion

Longitudinal Acceleration

The acceleration of a vehicle depends on the net force acting on it. The basic equation for longitudinal acceleration is:

$$F_{net} = mA \quad (1)$$

- Net force acting on the vehicle (N)
- m: Total mass of the vehicle (kg)
- A: Acceleration (m/s²)

3.1.2 Aerodynamic Drag

$$F_d = \frac{1}{2} C_d A_f \rho v^2 \quad (2)$$

Where:

- C_d : Coefficient of aerodynamic drag (dimensionless)
- A_f : Frontal area of the vehicle (m²)
- ρ : Air density (kg/m³)
- v: Speed of the vehicle (m/s)

3.1.3 Rolling Resistance

$$F_{zf}, F_{zr} = C_r mg \quad (3)$$

Where:

- C_r : Coefficient of rolling resistance (dimensionless)
- g: Acceleration due to gravity (9.81 m/s²)

3.1.4 Slope Resistance (or Gradient)

$$F_{slope} = mg \sin(\delta) \quad (4)$$

δ Angle of inclination of the slope (rad).

3.1.5 Vehicle Motion Equation (Sum of Forces)

The sum of all forces acting on a vehicle gives the net force:

$$F_{net} = F_{traction} - (F_d + F_{zf}, F_{zr} + F_{slope})$$

The vehicle model assumes the vehicle is moving in a straight direction under braking conditions, as shown in Figure 2. Equation (5) [25] represents the equilibrium for the horizontal direction:

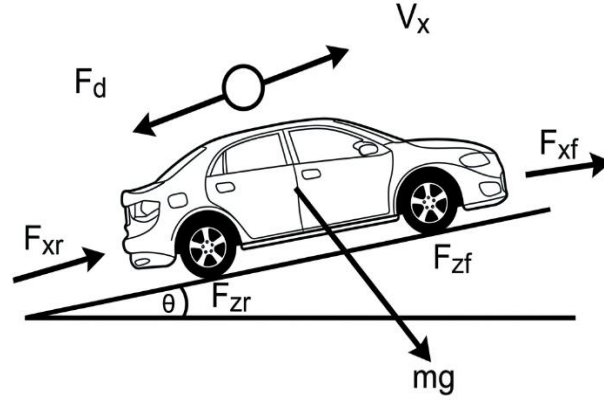


Figure 2: Vehicle Model variable

$$F_x = F_i \quad (5)$$

Where F_x is the friction force between the wheel and the ground, and F_i is the inertial force of the vehicle. The equation of equilibrium for the vertical direction is:

$$N = W \quad (6)$$

Where N is the normal force (road reaction) and W is the vehicle weight.

The friction force [26] can be expressed as:

$$F_x = \mu \times N \quad (7)$$

Where μ is the coefficient of friction between the wheel and the road.

The vehicle's weight is:

$$W = m \times g \quad (8)$$

By substituting equations (6) and (8) into equation (7), the expression for the friction force is:

$$F_x = \mu \times m \times N \quad (9)$$

Where mv is the vehicle mass and g is the acceleration due to gravity

Table 1: Electric vehicle dynamics variables.

Parameter	value
Mass1	600 Kg
Mass2	900 Kg
Mass3	1200 Kg
The quantity of wheels on each axle	2

Horizontal separation between the front axle and the CG	1.4 m
Horizontal separation between the rear axle and the CG	1.4 m
CG elevation above the ground	0.5
frontal region	1 m ²
coefficient of drag	0.26

4. PMSM Model and Optimization

The overall parameters of EVs, particularly the dimension's limits, rating, and peak indicators, and operating circumstances, are compared with the PMSM's needed performance standards [27]. The size equation of the PMSM is expressed as follows to finish the motor's primary size parameter design:

$$D_{il}^2 L_{ef} = \frac{P_{pout}}{(\pi \sqrt{2}/4) \omega B_{\delta} K_{dp} A'} \quad (10)$$

where L_{ef} is the core length, D_{il} is the stator inner diameter, P_{out} is the motor output power, is the rotor angular speed, B_{δ} is the air-gap flux density amplitude, $B_{\delta} A'$ is the electromagnetic load, K_{dp} is the winding coefficient, and A is the electrical load.

$$A = \frac{Z N_s I}{a \pi D_{il}} \quad (11)$$

where I is the phase current's effective value, Z is the number of stator slots, N_s is the number of conductors per slot, and a is the number of parallel branches.

Equation (10) states that the speed and electromagnetic load of the PMSM influence its power density contribution. The development of a high-power-density motor is likely to occur throughout a wide speed range. In order to increase the weak magnetic depth, PMSM uses a "V-" magnetic pole structure. The motor's heating and temperature rise are directly impacted by the thermal load, which is stated as

$$Q = AJ \quad (12)$$

where J denotes the current density, and Q denotes the thermal load. In order to prevent a significant increase in temperature, the thermal load cannot be exceeded. To enhance the wire's cross-sectional area, a lower current density should be chosen, while increasing the electrical load improves power density. Thus, flat wire wrapping technology is used by PMSM [28].

A mathematical model that takes iron losses into account. The PMSM mathematical model takes into account iron losses established in the d-q coordinate system to study the internal operating mechanism of the high-speed, high-power-density motor with significant internal iron losses. Additionally, as Figure 3 illustrates, the iron loss in the motor is equal to the iron loss resistance

R_c and the induction electromotive force (EMF) of the d-q axis in parallel. The iron loss current (i_{dc} , i_{qc}) and torque current (i_{dt} , i_{qt}) are separated from the d-axis and q-axis stator current (i_d , i_q).

$$u_d = R_s i_d + L_d^{diff} \frac{di_{dt}}{dt} - \omega L_q i_{qt} \quad (13)$$

$$u_q = R_s i_q + L_q^{diff} \frac{di_{qt}}{dt} - \omega L_d i_{dt} + \omega \psi_f \quad (14)$$

where u_d and u_q are the stator d -axis and q -axis voltage; R_s is the winding resistance; L_d^{diff} and L_q^{diff} are differential inductances of d - and q -axis; L_d and L_q are full inductances of d - and q -axis; and ψ_f is the permanent magnet flux linkage.

Flux linkage equation can be expressed as follows:

$$\begin{cases} \psi_d = L_d i_{dt} + \psi_f \\ \psi_q = L_q i_{qt} \end{cases} \quad (15)$$

where ψ_d and ψ_q is the stator d -axis and q -axis Flux linkage. Electromagnetic torque is defined as and contains full inductances:

$$T_m = p(\psi_d i_{qt} - \psi_q i_{dt}) = p[(L_d - L_q)i_{dt}i_{qt} + \psi_f i_{qt}] \quad (16)$$

where T_m is the electromagnetic torque; p is the number of pole pairs.

The e input power of the PMSM is obtained from above mathematical model and contains full inductances

$$P_{in} = u_d i_d + u_q i_q = \frac{\omega^2}{R_c} [(L_d i_{dt} + \psi_f)^2 + (L_q i_{qt})^2] + R_s (i_d^2 + i_q^2) + \omega [(L_d - L_q)i_{dt}i_{qt} + \psi_f i_{qt}], \quad (17)$$

where P_{in} is the input power; and R_c is the motor iron loss resistance. The first part on the left is the iron loss P_{Fe} . The second part is the copper loss of the winding P_{Cu} . The third part is the motor electromagnetic power P_{em} , which includes mechanical loss P_f , stray loss P_s , and output power P_{out}

The PMSM efficiency is as follows

$$\eta = \frac{P_{out}}{P_{IN}} \times 100 \quad (18)$$

The salient pole ratio is introduced for optimal analysis

$$p = \frac{L_q}{L_d} \quad (19)$$

Equations (16) and (17) state that the reluctance torque and permanent magnet torque make up the electromagnetic torque produced by the salient pole PMSM. Reluctance torque is produced because the motor's d-axis inductance is less than its q-inductance, which biases the magnetic flux in the route selection. Increasing the motor's salient pole ratio can boost electromagnetic torque and efficiency while the current is constant, but it also raises the magnetic field's harmonic content. Multi objective optimization of the PMSM rotor structure is required because as shown in figure 3, while increasing the rotor permanent magnet flux can greatly enhance the output torque and power, its iron losses will also increase.

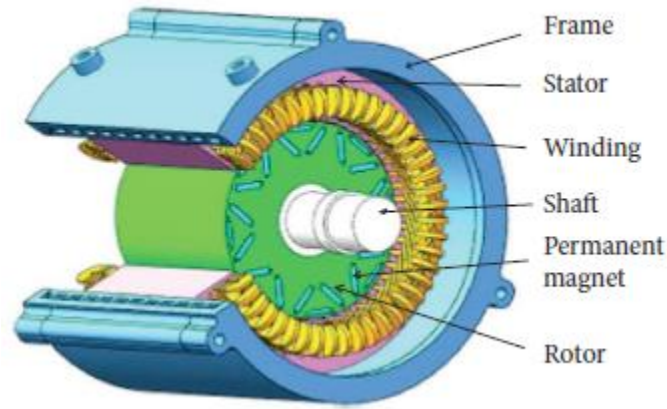


Figure 3. Construction of PMSM [36].

The integration of the optimization algorithm with the motor drive control structure is shown in the diagram of the field-oriented control system for permanent magnet synchronous motors in electric vehicles based on particle swarm optimization as shown in figure 4. The electric vehicle reference speed input, speed controller, current control loops, coordinate transformation blocks, PWM inverter, PMSM model, and PSO optimization unit are the usual components of the system. In order to improve speed response, decrease torque ripple, and increase energy efficiency, the PSO method is used to optimize controller settings such as proportional–integral gains. In order to reduce system error and enhance dynamic performance, the PSO algorithm iteratively modifies the control parameters while the PMSM feedback signals, such as speed and current, are continuously monitored and compared with reference values. For high-performance electric vehicle applications, the optimized control structure offers reliable motor operation, precise torque control, and economical energy use.

Using MATLAB/Simulink models as shown in figure 5, particle swarm optimization algorithms have been extensively used to optimize the field-oriented control of permanent magnet synchronous motors for electric vehicle applications. This method uses PSO to find the best control parameters that minimize torque ripple and power losses, improve motor performance, and increase speed tracking accuracy. Simulink is used to model the PMSM drive system in order to assess the motor's dynamic and steady-state behavior under various operating circumstances.

When PSO is combined with the FOC strategy, the controller outperforms traditional tuning techniques in terms of response time, stability, and energy efficiency. The PSO-optimized control system offers improved electromagnetic torque characteristics, more seamless speed regulation, and improved overall performance, according to simulation results, making it ideal for electric vehicle drive applications.

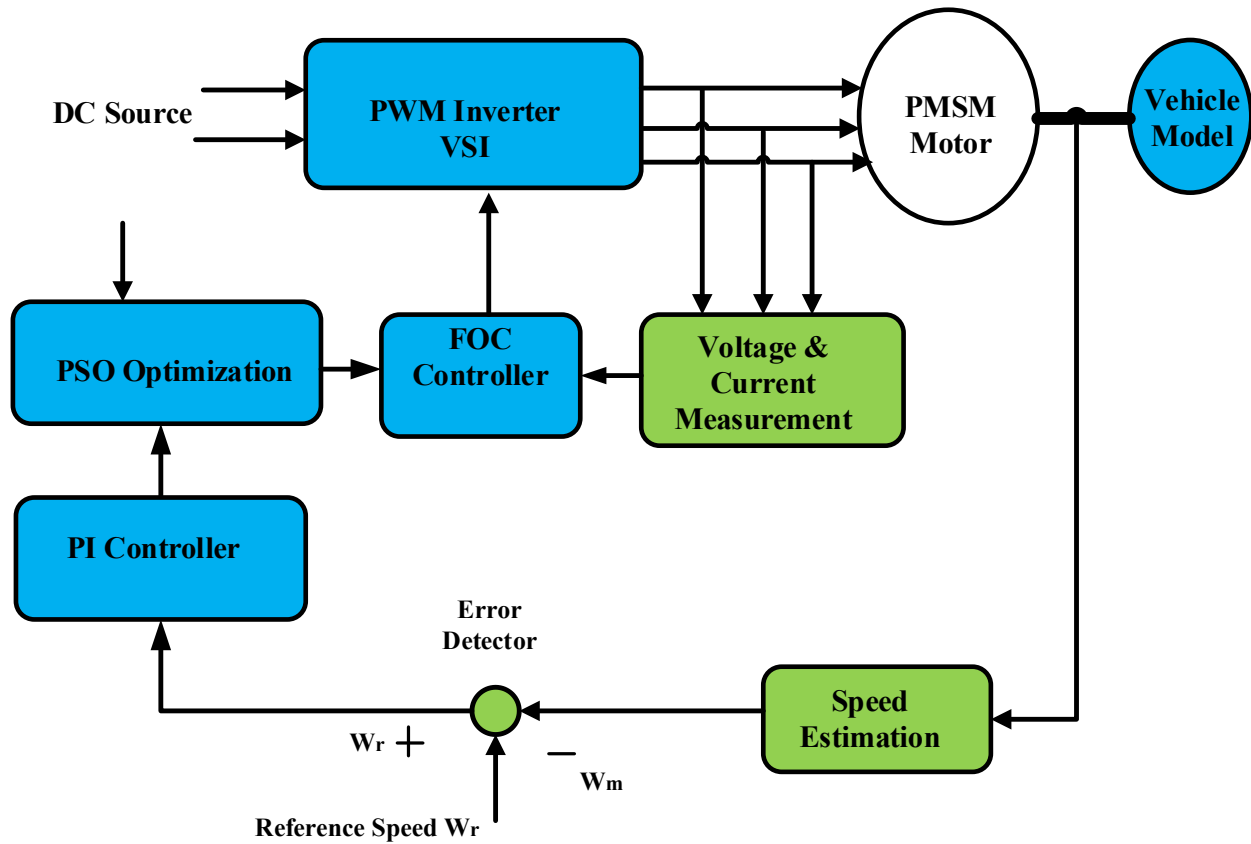


Figure 4. The optimization algorithms FOC Permanent Magnet Synchronous Motor for Electric Vehicles.

Table 2: Calculate the total energy consumed by the prototype electric vehicle for various total weights.

Description	Different Total Weight	Battery Energy	Acceleration	Deceleration	Speed Vehicle	
Mass1	600 Kg	4453 Joule	2 sec	2 sec	37 Km/h	26 Km/h
Mass2	900 Kg	4453 Joule	2 sec	2 sec	26 Km/h	18 Km/h
Mass3	1200 Kg	4453 Joule	2 sec	2 sec	20 Km/h	15 Km/h

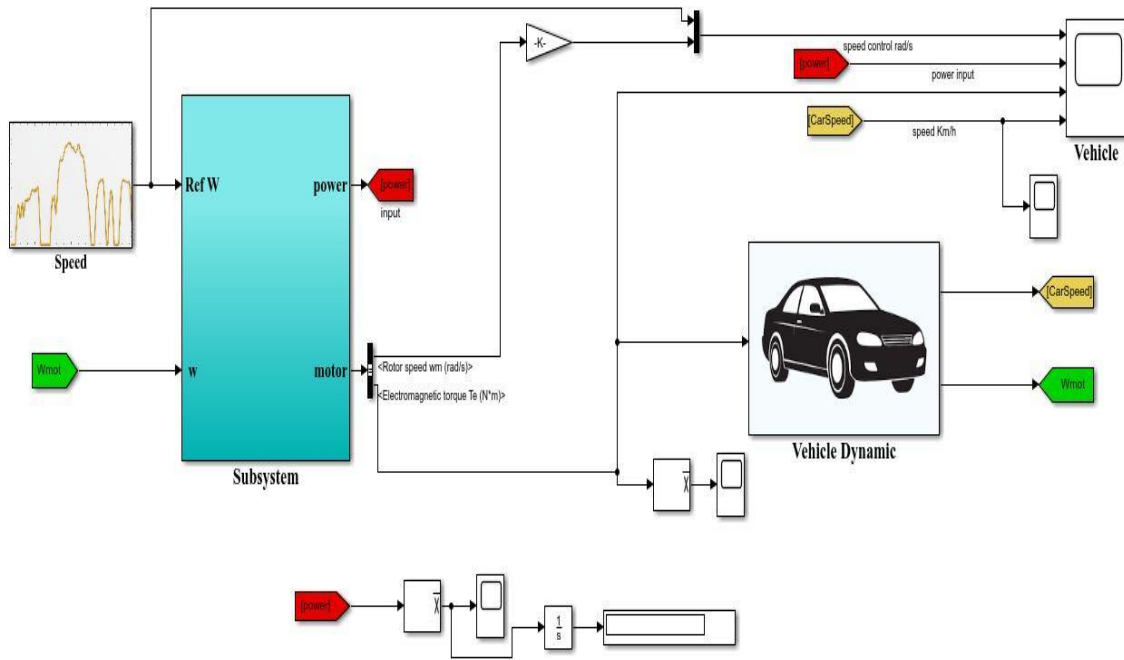


Figure 5: MATLAB/Simulink models PMSM for Electric Vehicles.

To assess the impact of load fluctuation on energy demand and overall system performance, the prototype electric vehicle's total energy consumption was computed for a range of total vehicle weights. Due to the increased torque and power needed for acceleration and motion, the results indicated that the overall energy consumption rose proportionately with the vehicle weight. While lighter loads led to increased efficiency and lower energy consumption, heavier vehicle circumstances increased current draw and battery energy use. The study showed that driving range, motor performance, and regenerative braking effectiveness are all significantly impacted by vehicle weight, underscoring the need of maximizing vehicle mass in electric vehicle design and energy management systems as shown in table 2.

5. Simulation Results

The control algorithm created during this work was tested and analyzed using a 600 kg vehicle in a simulation environment. Models suitable for the electrical machine, power supply, and other powertrain components should be included in the simulation scenario. Figure 6 displays the electric vehicle's mechanical speed, and displays the Permanent Magnet Synchronous Motor's speed (rpm), Figure 7 displays the motor's stator phase current at different speeds, and the Permanent Magnet Synchronous Motor's stator line voltage at different speeds.

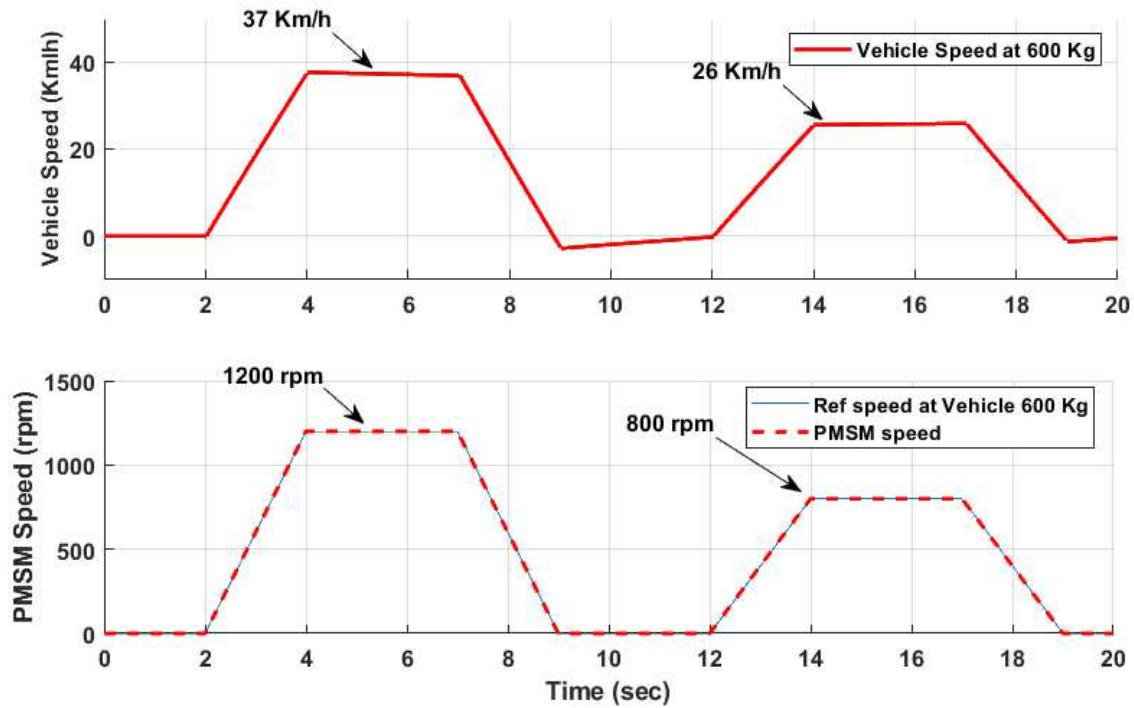


Figure 6: Displays the electric vehicle's mechanical speed, and displays the Permanent Magnet Synchronous Motor's speed (rpm) at vehicle 600 Kg

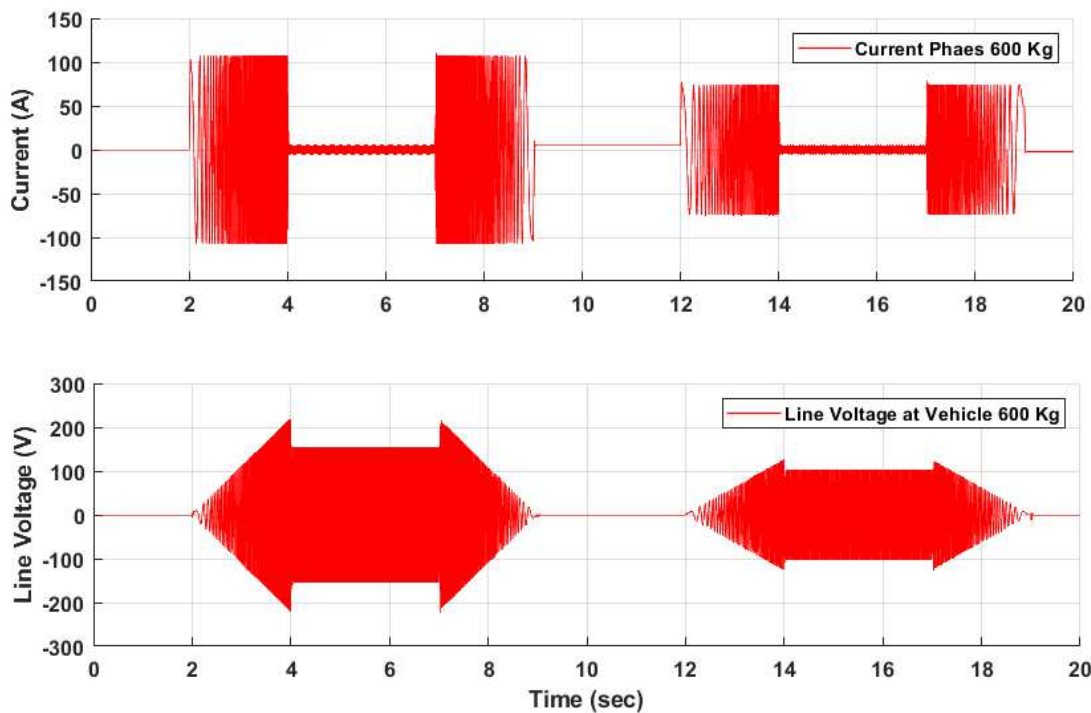


Figure 7: Displays the motor's stator phase current at different speeds, and the Permanent Magnet Synchronous Motor's stator line voltage at different speeds at vehicle 600 Kg

The control algorithm created during this work was tested and analyzed using a 900 kg vehicle in a simulation environment. Models suitable for the electrical machine, power supply, and other powertrain components should be included in the simulation scenario. Figure 8 displays the electric vehicle's mechanical speed, and displays the Permanent Magnet Synchronous Motor's speed (rpm), Figure 9 displays the motor's stator phase current at different speeds, and the Permanent Magnet Synchronous Motor's stator line voltage at different speeds.

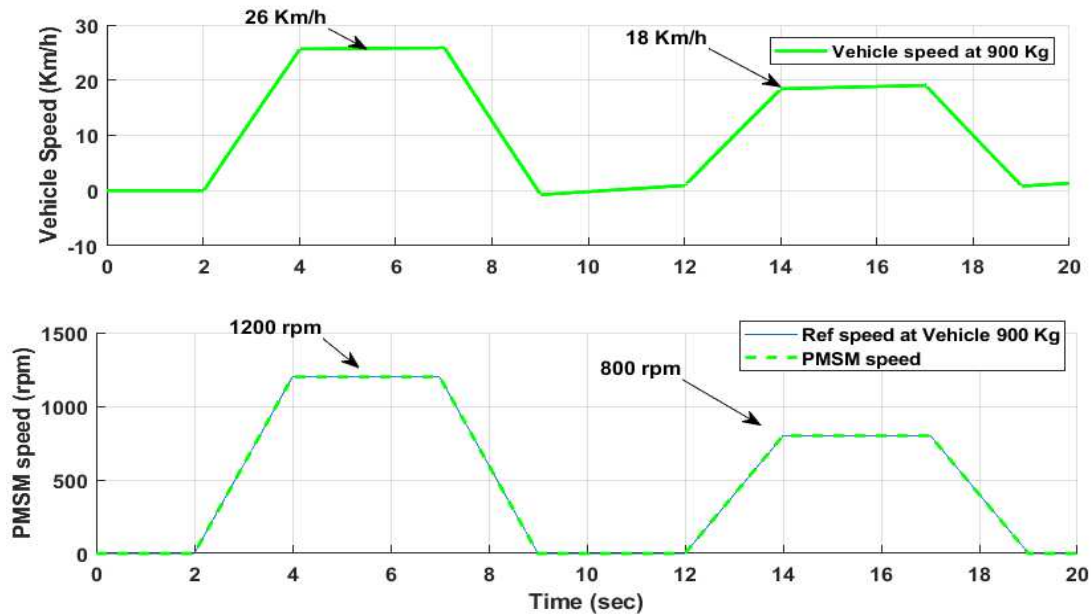


Figure 8: Displays the electric vehicle's mechanical speed, and displays the Permanent Magnet Synchronous Motor's speed (rpm) at vehicle 900 Kg

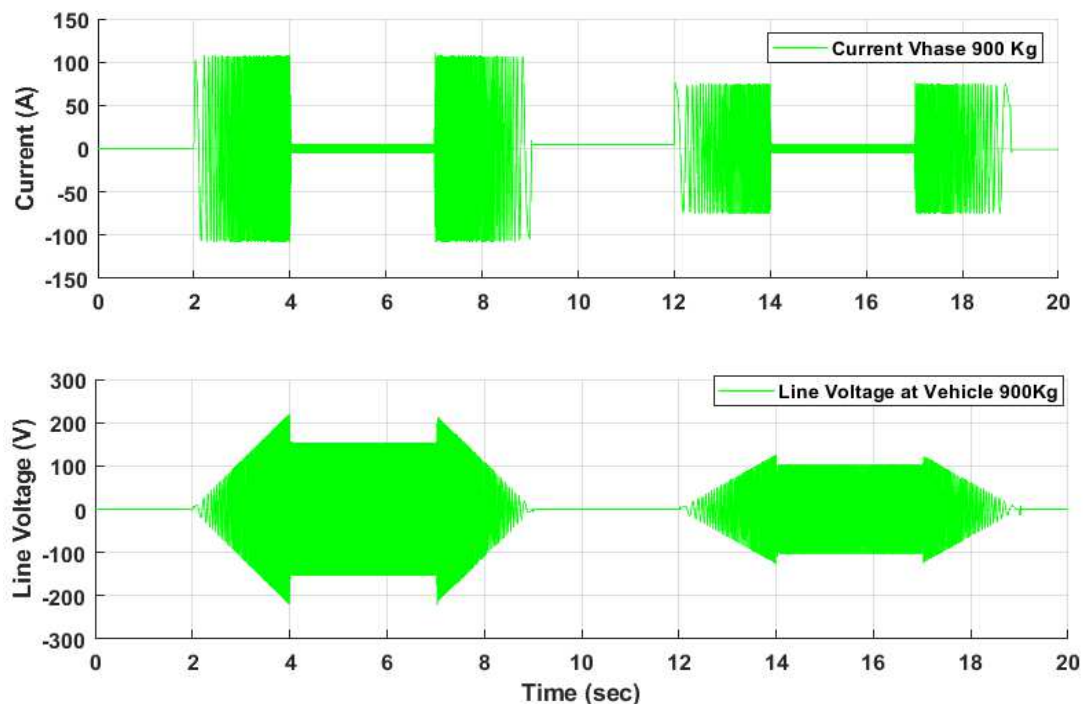


Figure 9: Displays the motor's stator phase current at different speeds, and the Permanent Magnet Synchronous Motor's stator line voltage at different speeds at vehicle 900 Kg

The control algorithm created during this work was tested and analyzed using a 1200 kg vehicle in a simulation environment. Models suitable for the electrical machine, power supply, and other powertrain components should be included in the simulation scenario. Figure 10 displays the electric vehicle's mechanical speed, and displays the Permanent Magnet Synchronous Motor's speed (rpm), Figure 11 displays the motor's stator phase current at different speeds, and the Permanent Magnet Synchronous Motor's stator line voltage at different speeds.

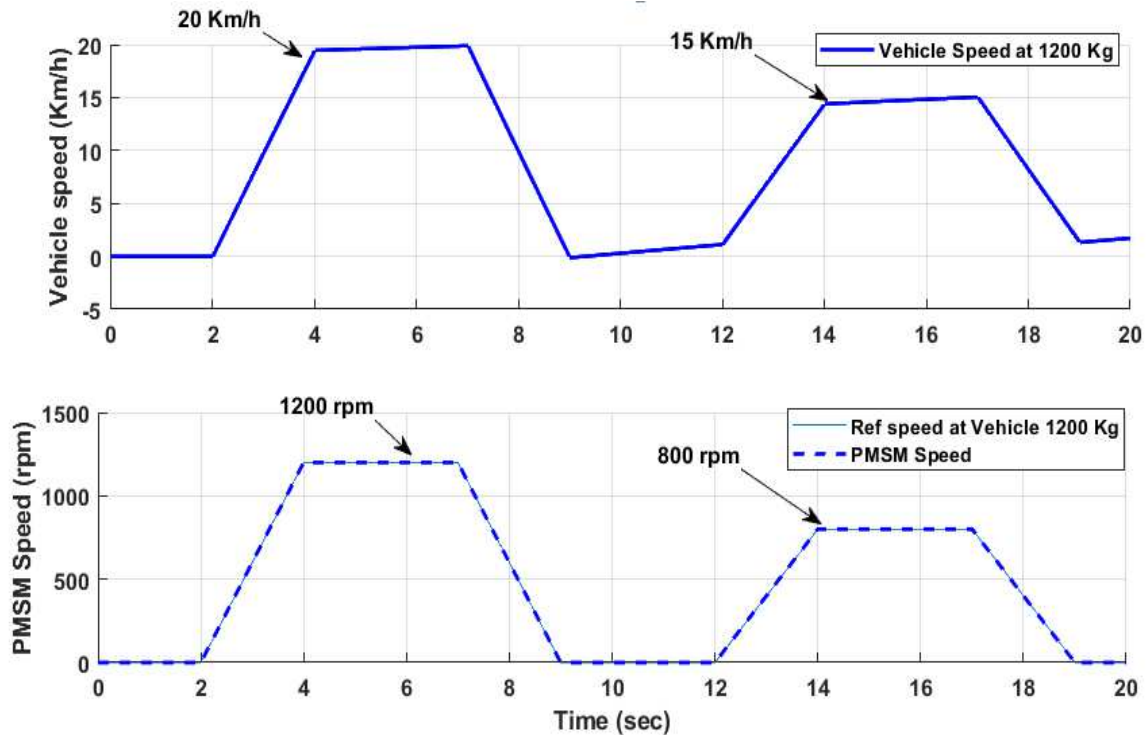


Figure 10: Displays the electric vehicle's mechanical speed, and displays the Permanent Magnet Synchronous Motor's speed (rpm) at vehicle 1200 Kg

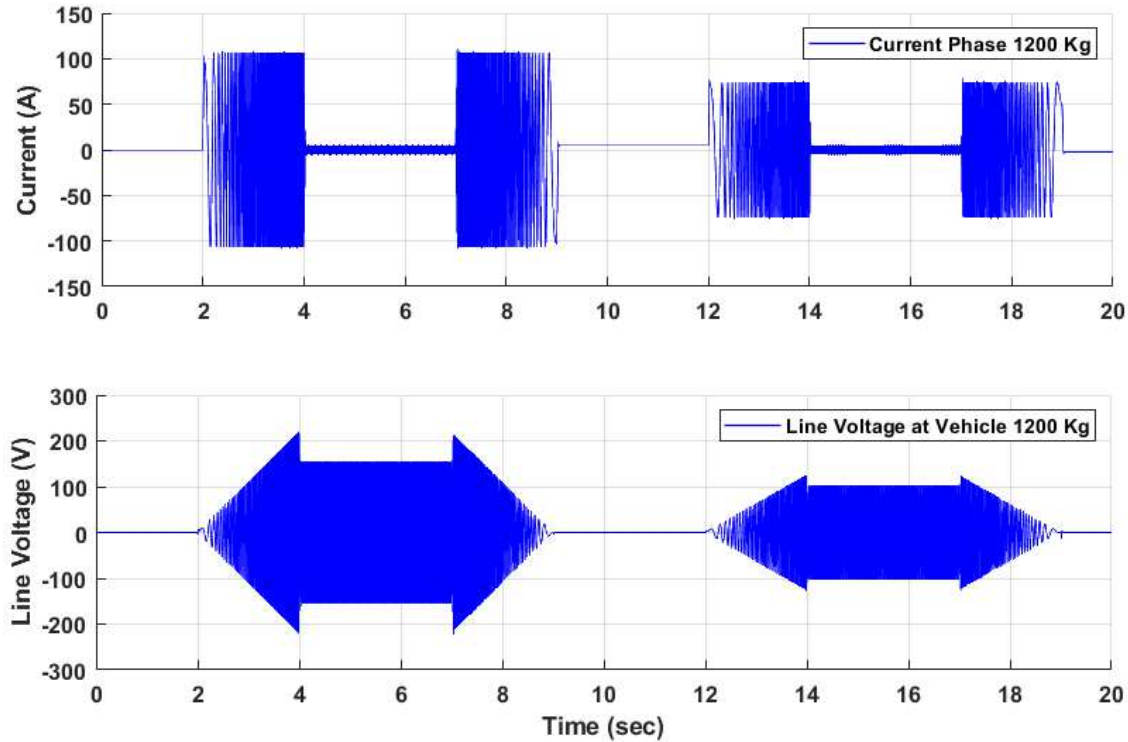


Figure 11: Displays the motor's stator phase current at different speeds, and the Permanent Magnet Synchronous Motor's stator line voltage at different speeds at vehicle 1200 Kg

6. Comparison Between Different Total Weight

The prototype electric vehicle's performance was examined in a range of total weight scenarios with the same amount of energy and varying speeds as shown table 3. The purpose of the study was to find out how vehicle mass and speed variation affected motor performance, energy consumption, and system efficiency. The vehicle's dynamic reaction and power demand were shown to be greatly impacted by changes in overall weight. However, keeping the energy level constant made it possible to assess how well the drive system handled various loading scenarios. The findings showed that energy consumption, the effectiveness of regenerative braking, and the overall efficiency of electric vehicle operation are all significantly influenced by vehicle weight and speed.

Figure 12 illustrates the torque under mechanical speed of the electric vehicle, and the power from the DC battery of the PMSM under various speeds. Weights ranging from 600, 9000, and 1200 kg, with the same power drawn from the source and the torque value remaining constant, resulted in a change in speed value.

Table 3: The various total weights while keeping the energy

Description	Different Total Weight	Speed Vehicle	Full Speed	Speed Vehicle	Full Speed
Mass1	600 Kg	37 Km/h	100%	26 Km/h	100%
Mass2	900 Kg	26 Km/h	70.27%	18 Km/h	69.23%
Mass3	1200 Kg	20 Km/h	54.05%	15 Km/h	57.69%

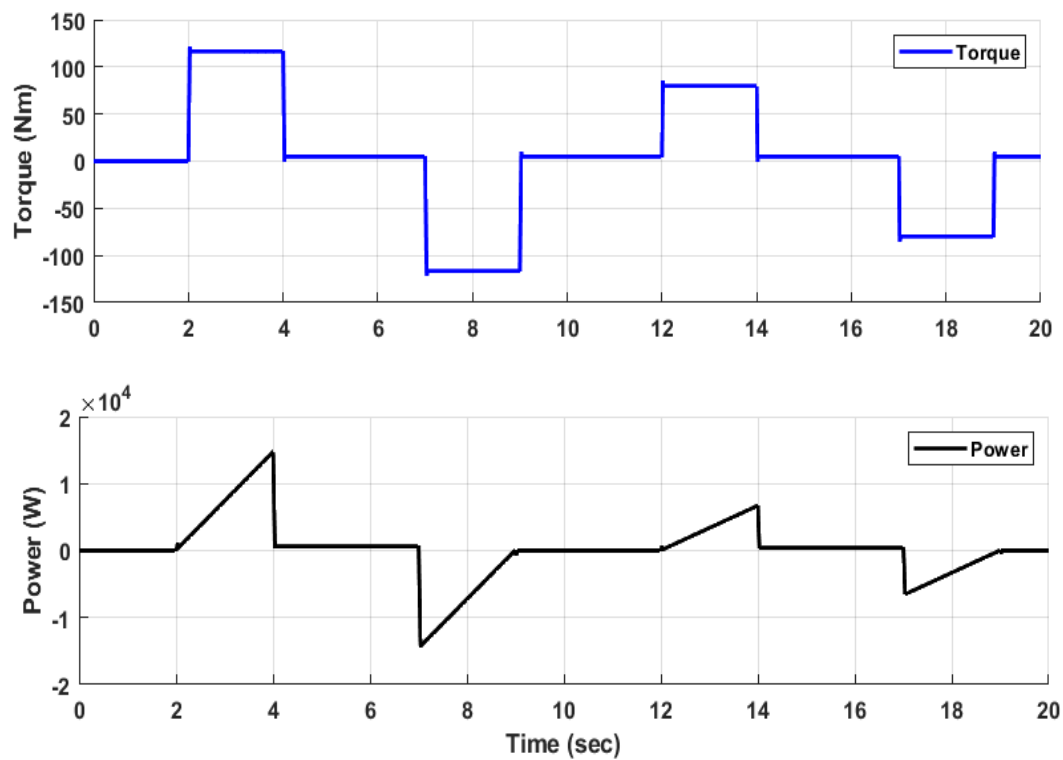


Figure 12: The torque under mechanical speed of the electric vehicle, and the power from the battery of the PMSM under various speeds.

To assess the impact of mass variation on dynamic performance, the vehicle speed was examined under various vehicle weight situations while keeping the same energy level of 4453 joules. The findings demonstrated that, under the same available energy conditions, lighter vehicles outperformed heavier ones in terms of speed. This pattern is explained by the fact that lighter vehicles require less power and inertia to accelerate. On the other hand, because more force and torque were required to overcome the extra mass, increasing the overall weight of the vehicle resulted in a decrease in the speed that could be attained. The analysis highlights the significance of weight reduction in enhancing electric vehicle economy and driving capabilities by showing a

clear correlation between vehicle weight and speed performance at constant energy levels as shown in figure 13,14.

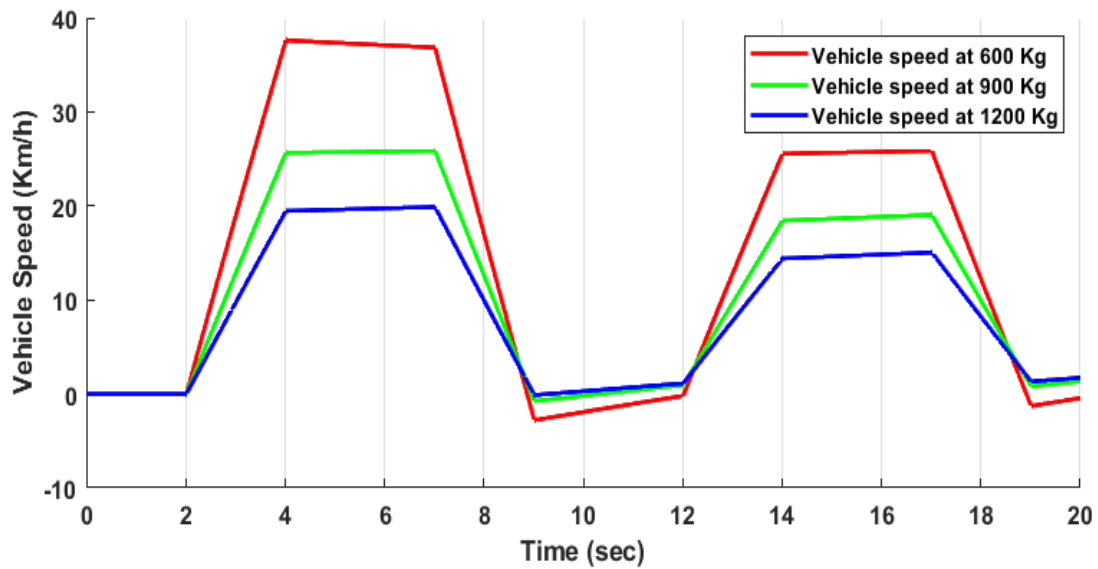


Figure 13: Vehicle speed at various weights with the same 4453 joules of energy

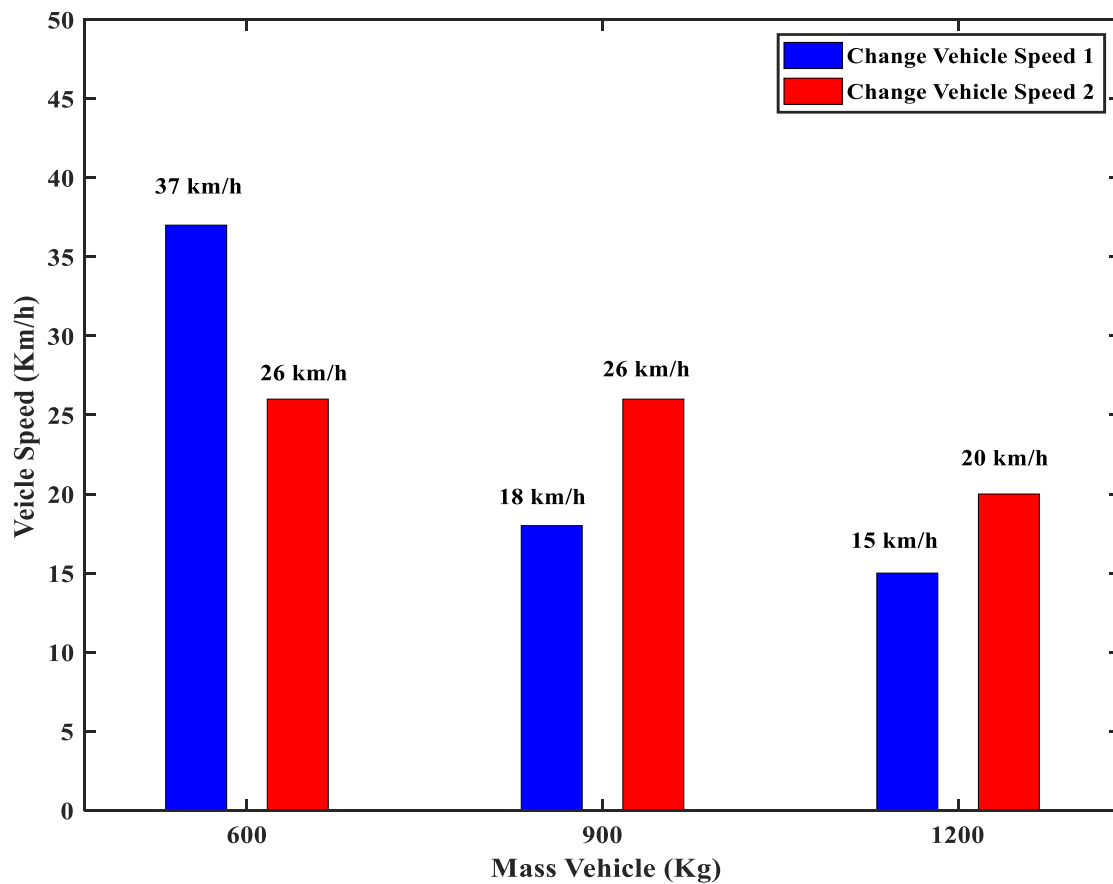


Figure 14: Vehicle speed at different weight vehicles

7. Conclusion

To assess the impact of mass variation on dynamic performance, the vehicle speed was examined under various vehicle weight situations while keeping the same energy level of 4453 joules. The findings demonstrated that, under the same available energy conditions, lighter vehicles outperformed heavier ones in terms of speed. This pattern is explained by the fact that lighter vehicles require less power and inertia to accelerate. On the other hand, because more force and torque were required to overcome the extra mass, increasing the overall weight of the vehicle resulted in a decrease in the speed that could be attained.

The 600 kg electric vehicle's initial driving cycle recorded speeds of 37 km/h and 26 km/h, according to the PMSM authorized driving cycle. The fastest speed of 26 km/h and 18 km/h was attained by the 900 kg vehicle. Based on the electric motor's acceleration and consistent battery power consumption, the 1200 kg vehicle had the slowest speed, hitting 20 km/h and 15 km/h, respectively.

When a permanent magnet synchronous motor was used in place of an induction motor, lower values of drawn power were observed at various operating speeds in contrast to Research [37]. The PMSM drive system's improved torque characteristics, lower rotor losses, and increased efficiency are the primary reasons for this improvement. PMSM eliminate copper losses and improve overall energy conversion efficiency because they don't require rotor excitation current, unlike induction motors. Consequently, under different speed situations, the PMSM showed enhanced dynamic performance and reduced power consumption. These results suggest that permanent magnet synchronous motors are better suited for electric vehicle applications that need high efficiency, lower energy consumption, and improved performance when operating at varied speeds.

Declarations

Ethical approval: NOT APPLICABLE

Consent to participate: YES

Consent to publish: YES

Declaration of Conflicts of Interest

Funding: The Authors state that this research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Conflicts of interest/Competing interests: Nil

Data availability: Yes. I used or generated research data in this study

Availability of data and material: Data files are available upon request

Clinical trial: NOT APPLICABLE

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